Difference of Two Squares

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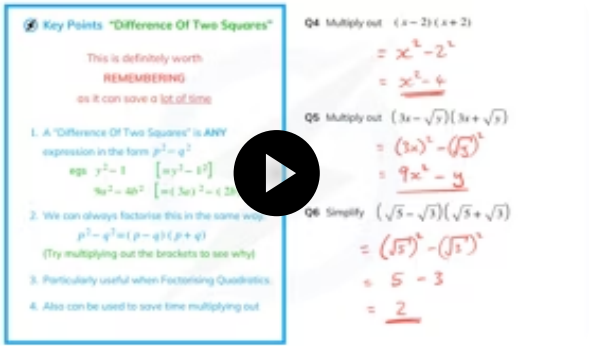

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Difference of two squares

What is the difference of two squares?

- When a "squared" quantity is subtracted from another "squared" quantity, you get the **difference** of **two squares**
  - For example:
    - $a^2 - b^2$
    - $9^2 - 5^2$
    - $(x + 1)^2 - (x - 4)^2$
    - $4m^2 - 25n^2$, which is $(2m)^2 - (5n)^2$

How do I factorise the difference of two squares?

- **$a^2 - b^2$ factorises to $(a + b)(a - b)$**
  - This can be shown by expanding the brackets
    - $(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2$
  - The brackets can swap order
    - **$a^2 - b^2 = (a + b)(a - b) = (a - b)(a + b)$**
    - (but terms inside a bracket cannot swap order)
- For example, $x^2 - 9 = (x+3)(x-3)$
  - This **is the same** as $(x-3)(x+3)$
  - But **not the same** as $(3+x)(3-x)$
    - which expands to $9 - x^2$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- You may find it used with **numbers**
  - $7^2 - 3^2 = (7+3)(7-3) = (10)(4) = 40$
- You can use a combination of **square numbers** and **squared variables**
  - $4m^2 - 9n^2 = (2m)^2 - (3n)^2 = (2m + 3n)(2m - 3n)$
- You can use **other powers** which can be written as a **difference of two squares**
  - $r^8 - t^6 = (r^4)^2 - (t^3)^2 = (r^4 + t^3)(r^4 - t^3)$
- You can use the difference of two squares **more than once** to **fully factorise** expressions
  - $a^4 - b^4 = (a^2)^2 - (b^2)^2 = (a^2 + b^2)(a^2 - b^2) = (a^2 + b^2)(a + b)(a - b)$
- You may also need to **take out a common factor first**
  - $2x^2 - 18 = 2(x^2 - 9)$ giving $2(x + 3)(x - 3)$
    - The 2 comes out in front

Can I use the difference of two squares to expand?

- Using the difference of two squares to **expand** is **quicker** than expanding double brackets and collecting like terms
- Brackets of the form **$(a + b)(a - b)$ expand to $a^2 - b^2$**
  - For example $(2x + 3)(2x - 3)$ expands to $4x^2 - 9$

Examiner Tips and Tricks

The difference between two squares is often the "trick" required to complete a harder algebraic question in the exam.

Learning to spot it can help speed up your working.

Worked Example

(a) Factorise $9x^2 - 16$.

Answer:

Recognise that $9x^2$ and 16 are both squared terms

Therefore you can factorise using the difference of two squares

Rewrite as a difference of two squared terms

$$9x^2 - 16 = (3x)^2 - (4)^2$$

Use the rule $a^2 - b^2 = (a + b)(a - b)$

$$(3x + 4)(3x - 4)$$

(b) Factorise $4r^2 - t^4$.

Therefore you can factorise using the difference of two squares

Rewrite as a difference of two squared terms

$$4r^2 - t^4 = (2r)^2 - (t^2)^2$$

Use the rule $a^2 - b^2 = (a+b)(a-b)$

$$(2r + t^2)(2r - t^2)$$

(c) Factorise $2y^2 - 50$

Answer:

This does not appear to be in the form $a^2 - b^2$

There is a common factor of 2, so take this factor out

$$2(y^2 - 25)$$

You can now see $y^2 - 25$ which has the form $y^2 - 5^2$

Use the rule $a^2 - b^2 = (a+b)(a-b)$

$$y^2 - 25 = (y+5)(y-5)$$

Now multiply this answer by 2 (leaving the 2 on the outside)

$$2(y+5)(y-5)$$

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